Daily Problem November 22, 2025

Find:

$$\int \frac{1}{1+x^4}dx.$$

Solution:

$$\frac{1}{1+x^4} = \frac{1}{2}\left(\frac{1+x^2}{1+x^4} + \frac{1-x^2}{1+x^4}\right) \stackrel{\text{def}}{=} \frac{1}{2}\left(I_1 + I_2\right)$$

$$I_1 = \frac{1+x^2}{1+x^4} = \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} = \frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} = \frac{1}{\sqrt{2}}\arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)$$

In a similar way, we can get:

$$I_2 = \frac{1-x^2}{1+x^4} = \frac{\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}} = -\frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2} = \frac{1}{2\sqrt{2}}\ln\left|\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right|$$

In conclusion:

$$\int \frac{1}{1+x^4}dx = \frac{1}{2}\left(I_1+I_2\right) = \frac{1}{4\sqrt{2}}\ln\left|\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right| + \frac{1}{2\sqrt{2}}\arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right) + C$$

For interested students, you can try to compute:

$$\int \frac{1}{1+x^5}dx$$